



Léry Monnerat

Master of Engineering (ENSTA PARIS) - Master of Science (Applied Mathematics - Optimization)

Spécialization in Applied Mathematics

📅 20/11/2003
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🚗 permis B

Education

Master 2 - Paris-Saclay University, Saclay

📅 2025 - 2026

Optimization. Double degree with the 3rd year at ENSTA Paris.

Advanced coursework in Optimization and Operations Research, Optimal Control and Decision Processes, Machine Learning and Complexity, including Discrete Optimization, Integer Optimization for Machine Learning, Derivative-Free Optimization, Optimal Control of ODEs, Markov Decision Processes, Tropical Algebra and Games, Deep Learning, and Complexity Theory.

Master 1 - Institut Polytechnique de Paris, Saclay

📅 2024 - 2025

Applied Mathematics. Double degree with the 2nd year at ENSTA Paris.

Functionnal analysis, spectral theory, Analysis and finite element approximation of PDEs, Probability (Markov chains, discrete time martingales), statistical modeling, Optimization, system control, operation research, Scientific calculation (numerical simulation, hpc, numerical linear algebra).

Master of Engineering - ENSTA Paris, Palaiseau

📅 2023 - 2026

France's top 5 ranked school for science and engineering, founding member of the Polytechnic Institute of Paris
Mathematics, Computer Science, Physics, Economics, Management/Project Management, Law, Languages, Sports.
Specialization in **Applied Mathematics**.

Classes préparatoires - Lycée Marcelin Berthelot, Saint-Maur-des-Fossés

📅 2021 - 2023

2-year intensive program to prepare for the nationwide competitive entrance exams to top French engineering schools (first two years of a bachelor degree : Mathematics, Physics-Chemistry, Computer Science, Engineering Science)

Lycée - Lycée Étienne Bezout, Nemours

📅 2018 - 2021

High school/secondary diplomas and certificates with highest honour.

Experience

Master's thesis - University of Padua

📅 May-October 2026

The Shrödinger system and stability of entropic optimal transport, under the supervision of Professor Giovanni Conforti.

Research internship - Institut de Physique du Globe de Paris

📅 June-August 2025

The aim of this research project was to assess the potential of a new passive seismic imaging technique for the subsurface.

Workman internship - Village potager, Saint Pierre lès Nemours

📅 July 2024

Packaging, order preparation, market gardening.

Citizen engagement

📅 2023 - 2025

- Leading of a robotics course ("Semaine de la Robotique") for secondary school students. 📍 Palaiseau
- Cordées de la réussite : creation and running of scientific and cultural activities for middle school students. 📍 Trappes

Skills

Computer science

Bash C/C++ Python Matlab $\LaTeX$ Linux GIT

Projects (some)

C	Reinforcement learning (Q-learning, Sarsa).
C++	Traveling Salesman Problem (genetic algorithms).
Matlab	Rayleigh-Bénard Convection (fluid mechanics).
TIPE	Granular media and the urban construction supply chain (mechanics - mathematics)

🔗 lerymonnerat/IN104_ML
🔗 gabrielvignon2003/projet-VDC
🔗 lerymonnerat/MO102
🔗 lerymonnerat/tipe

Languages

French	Native
English	C1+
Spanish	A2/B1
Italian	A0/A1

Divers

Sports : competitive practice of ultimate frisbee (French champion in the U14 category).